Femtoscopic Study Using pA and AA Collsions at J-PARC

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Abstract

The short-range part of the baryon-baryon interaction, particularly the ΛN interaction, plays a crucial role in understanding high-density nuclear matter and the hyperon puzzle in neutron stars. While the Λ binding energy of ${}^{5}_{\Lambda}$He constrains the overall $\Lambda\alpha$ interaction, the short-range behavior remains poorly determined and model-dependent. We propose to study the $\alpha\Lambda$ momentum correlation in pA collisions at J-PARC (E88 experiment) and AA collisions at STAR Beam Energy Scan (BES) to probe this short-range interaction. Small source sizes ($\approx$ 1 fm) in pA collisions provide high sensitivity to the short-range part of the potential. Furthermore, we discuss future prospects for Λp femtoscopy at the J-PARC Heavy-Ion (J-PARC-HI) project, where polarization measurements will enable the decomposition of the interaction into its spin-singlet and triplet components.

Keywords: J-PARC, STAR, Femtoscopy, ΛN interaction, Hyperon Puzzle, J-PARC-HI

1. Introduction

The investigation of the baryon-baryon (BB) interaction, in particular the Λ-nucleon (ΛN) interaction at high densities, remains a cornerstone of nuclear and hadron physics. This research is fundamentally linked to the "hyperon puzzle" in the astrophysical community. The introduction of hyperons into dense matter is theoretically expected to soften the nuclear equation of state (EoS), which would limit the maximum mass of neutron stars. However, recent observations of heavy neutron stars with masses around $2M_{\odot}$ challenge this expectation, suggesting that a significant repulsion must exist in the high-density BB interaction. Despite its importance, providing experimental constraints on the short-range part of this interaction is a major challenge.

Historically, information on the ΛN interaction has been primarily derived from hypernuclear spectroscopy [1]. While hypernuclear spectroscopic data have successfully constrained the overall binding energies and bulk properties of hypernuclei, they often exhibit a limited sensitivity to the short-range details of the interaction. Multiple potential models with different internal structures can frequently reproduce the same binding energy, leading to a theoretical degeneracy.

Recently, significant progress has been made in scattering experiments. At J-PARC, the E40 experiment has successfully utilized a Σ^- beam to produce high-precision results on ΣN scattering [2, 3, 4]. Building on this success, new high-statistics scattering experiments using Λ beams are being planned at both J-PARC and SPring-8 [5]. Of particular interest is the J-PARC E86 experiment [6], which aims to measure Λp scattering in the 300–500 MeV/c momentum range. By measuring the depolarization of the Λ, E86 seeks to perform a full spin-decomposition, providing crucial information on the phase-shifts in the medium-energy (short-range) region for the two-body interaction in vacuum.

Furthermore, the J-PARC E90 experiment [7, 8] focuses on the precise spectroscopy of the ΣN cusp using the $d(K^-, \pi^-)$ reaction. By analyzing the cusp shape, E90 aims to determine the ΣN scattering length. The imaginary part of this scattering length is particularly significant as it directly reflects the coupling constant between the ΣN and ΛN channels. Since this coupling is expected to be large, the E90 experiment provides a critical and complementary insight into the baryon-baryon interaction, further enriching our understanding of the "hyperon puzzle" in neutron star cores.

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In this context, femtoscopy has emerged as a revolutionary tool that complements these traditional and spectroscopic methods. By measuring the momentum correlation between two particles at low relative momentum, femtoscopy provides a unique way to probe the space-time characteristics of the emission source and the underlying final-state interactions. Recent results from ALICE [9] and STAR [10] have demonstrated the power of this technique.

This paper discusses these complementary experimental programs at J-PARC. First, the J-PARC E86 and E90 experiments probe the short-range phase shifts and channel couplings. Second, we focus on $\alpha\Lambda$ femtoscopy (E88), which utilizes the high central density of the α particle to probe the short-range interaction *in-medium*. Third, we discuss the future J-PARC Heavy-Ion (J-PARC-HI) project, which aims to perform Λp femtoscopy with spin-decomposition to determine overall interaction parameters (scattering length and effective range) for each spin state in vacuum. These programs form a consistent and complementary research framework, spanning short-range to overall interactions in both vacuum and dense nuclear media.

2. $\alpha\Lambda$ Femtoscopy to probe short-range interaction

The $\alpha\Lambda$ system provides a unique platform to investigate the ΛN interaction. A significant constraint on this system comes from the well-established ground-state binding energy of the ${}^{5}_{\Lambda}$He hypernucleus ($B_\Lambda = 3.12 \pm 0.02$ MeV) [1]. Traditionally, hypernuclear spectroscopic data have been the primary source of information for determining the baryon-baryon interaction. However, as shown in Fig. 1, multiple theoretical potential models with drastically different spatial structures, such as isle-type and Nijmegen-type potentials, can equivalently reproduce the experimental binding energy. This degeneracy highlights that hypernuclear spectroscopy alone, while essential, is insufficient to uniquely determine the short-range part of the ΛN interaction.

The detailed behavior of the short-range ΛN interaction, including repulsive cores and three-body force effects (which can be understood as medium effects), is one of the most critical factors in addressing the "hyperon puzzle" in neutron stars. The maximum mass of a neutron star and the question of whether hyperons like Λ appear in its core depend strongly on the strength of the repulsion and the density-dependent aspects of the interaction at high densities. Moving beyond the bulk properties provided by hypernuclei requires a tool sensitive to the short-distance dynamics.

As demonstrated by Jinno *et al.* [11], femtoscopy with small source sizes ($R \lesssim 1.5$ fm) is an exceptionally powerful tool for this purpose. The resulting momentum correlation functions are highly sensitive to the differences in potential models, even when those models yield the same binding energy (see Fig. 2). In particular, proton-nucleus (pA) collisions at J-PARC energies naturally produce these small source sizes, offering a unique opportunity to provide definitive experimental constraints on the short-range and medium-dependent parts of the baryon-baryon interaction.

The use of α particles as a probe offers additional physical advantages. While most nuclei exhibit a "saturation density" $\rho_0 \approx 0.17$ fm^{-3} regardless of the atomic number, α particles are exceptionally dense, with a central density reaching nearly $2\rho_0$. This high-density environment allows for the study of the ΛN interaction at densities directly relevant to the core of neutron stars. Furthermore, while common femtoscopic pairs like Λp or $K^- p$ necessarily measure a weighted average of multiple spin ($S = 0, 1$) or isospin ($I = 0, 1$) states, the α particle has both zero isospin ($I = 0$) and zero spin ($J = 0$). Consequently, in femtoscopy, the $\alpha\Lambda$ system is in a unique spin and isospin state, providing a "clean" measurement of the interaction without the need for state decomposition.

3. Experimental studies at J-PARC and STAR

3.1. J-PARC E88 experiment

The J-PARC E88 experiment, also referred to as the Saϕre experiment, is designed to study the in-medium modification of the ϕ meson mass and width in nuclear matter. The primary goal is to investigate the partial restoration of chiral symmetry, which is expected to result in a reduction of the $\bar{s}s$ quark condensate at finite nuclear density. Unlike previous experiments that focused on the dilepton (e^+e^-) decay channel, E88 focuses on the high-statistics measurement of the $\phi \to K^+K^-$ channel in $p + A$ collisions at 30 GeV using the J-PARC primary proton beam.

A unique feature of E88 is the ability to study the momentum dependence of the mass shift (dispersion relation) and to distinguish between the longitudinal and transverse polarization modes of the ϕ meson. Such detailed information is crucial for testing various theoretical models, including QCD sum rules, which predict different in-medium behaviors

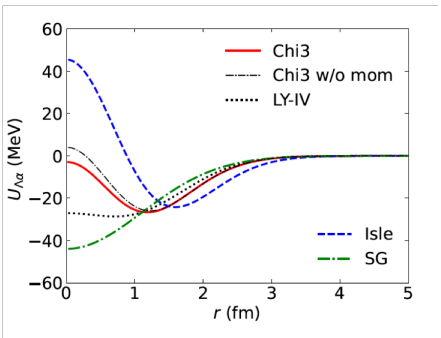

Figure 1: $\alpha\Lambda$ potentials for various models as functions of the distance between Λ and α. The phenomenological potentials in Gaussian form include Isle (dashed line) and SG (thick dash-dotted line), while the Skyrme-type potentials include Chi3 (solid line), LY-IV (dotted line), and Chi3 w/o mom (thin dash-dotted line). Details are referred to Ref. [1] and this figure is taken from Ref. [11].

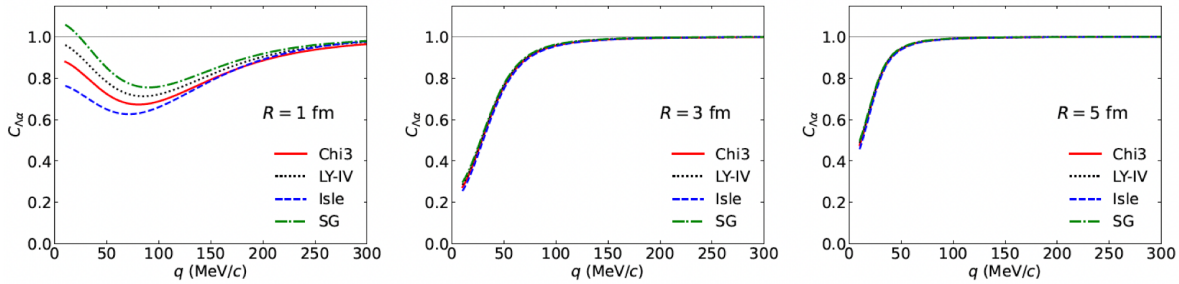

Figure 2: Predicted $\alpha\Lambda$ correlation functions for different source sizes and potential models. Taken from Ref. [11].

for different polarization states. To systematically study the density dependence, carbon (C), copper (Cu), and lead (Pb) targets will be used.

The J-PARC E88 experiment also serves as an essential benchmark for the future J-PARC Heavy-Ion (J-PARC-HI) project. While the full realization of the J-PARC-HI program is currently under planning with some uncertainties regarding the funding and timeline, the existing J-PARC accelerator complex already enables high-intensity pA collision experiments. E88 is therefore a critical milestone, demonstrating the feasibility of high-rate hadronic measurements. Beyond its primary focus on ϕ meson physics and $\alpha\Lambda$ femtoscopy, the experiment provides a versatile platform for various hadronic byproduct measurements, which will yield indispensable data for the design and optimization of the J-PARC-HI project.

The experiment utilizes a high-resolution magnetic spectrometer consisting of Gas Electron Multiplier (GEM) trackers, which are also used in the J-PARC E16 experiment [12]. For precision particle identification, the spectrometer is augmented with Multi-gap Resistive Plate Chambers (MRPC) for time-of-flight (ToF) measurements and threshold-type Aerogel Cherenkov counters to provide robust separation between kaons and pions up to several GeV/c. The layout of the spectrometer is shown in Fig. 3.

Notably, the wide acceptance and sufficient momentum resolution of this magnetic spectrometer allow for the successful detection of $\alpha\Lambda$ pairs as a byproduct. The Λ hyperons can be reconstructed via the $\Lambda \to p\pi^-$ decay. The α particles are identified by combining the magnetic rigidity and the specific energy loss (dE/dx) information obtained from the GEM trackers. The dE/dx measurement provides crucial information about the particle's charge (Z), enabling clear discrimination between $Z = 1$ particles (such as deuterons) and $Z = 2$ particles (such as ^{4}He), which is essential for the $\alpha\Lambda$ correlation analysis.

A critical challenge in femtoscopy is the simultaneous dependence of the correlation function on both the emission source size (R) and the final-state interaction parameters (scattering length a and effective range $r_{\rm eff}$). To uniquely extract the interaction, the source size must be precisely determined. In the E88 experiment, we aim to determine the individual source sizes of p, α, and Λ by analyzing the correlation functions of pp, $p\alpha$, and Λp pairs. The

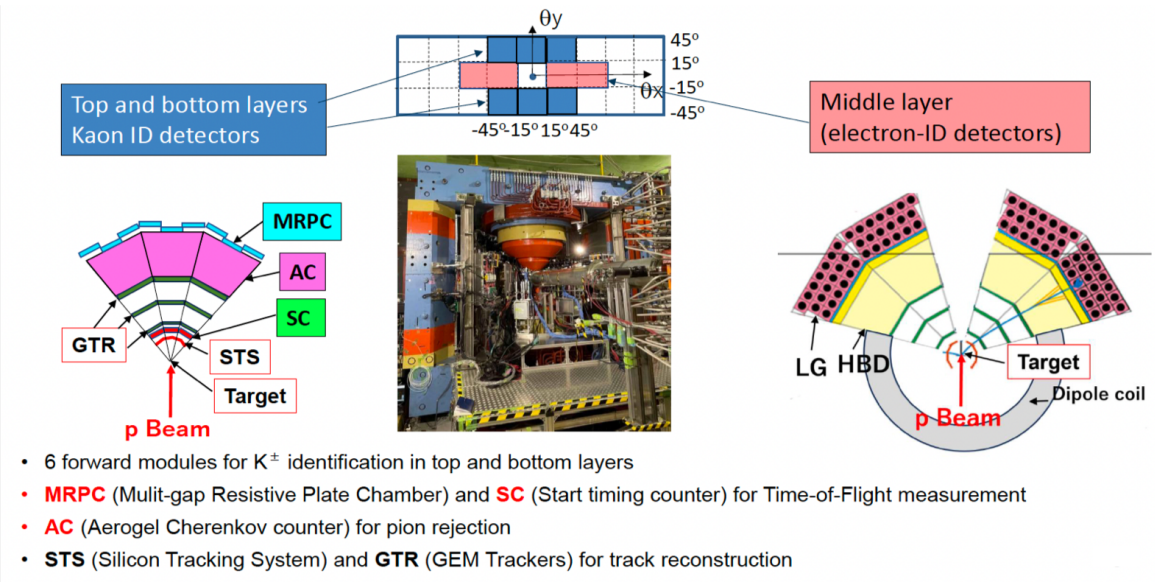

Figure 3: Experimental setup of the J-PARC E88 experiment.

interactions for pp and $p\alpha$ are well-established from extensive scattering data, and the Λp interaction is increasingly well-constrained from recent high-statistics femtoscopy at ALICE [9] and STAR [10], as well as from combined analyses of scattering and correlation data [13]. By fitting these pairs with the source size as the only free parameter, we can solve for the individual source sizes (R_p, R_α, R_Λ) as a system of equations, utilizing the known relationship that the pair source size is the geometric mean of the individual sizes. This allows us to accurately estimate the $\alpha\Lambda$ source size and thus provide the first experimental constraints on the short-range part of the $\alpha\Lambda$ interaction in pA collisions.

3.2. STAR Beam Energy Scan (BES)

To verify the consistency of the interaction models across different system sizes and energies, we also plan to analyze $\alpha\Lambda$ correlations in heavy-ion (AA) collisions. Specifically, we focus on the STAR Beam Energy Scan (BES) program at RHIC, particularly at $\sqrt{s_{NN}} \sim 7.7$ GeV. While the source size in AA collisions is generally larger than in pA, it is possible to explore smaller source sizes by selecting appropriate centrality classes. However, peripheral collisions introduce additional challenges; the source shape becomes significantly elliptical due to the initial overlap geometry. This non-spherical shape means that the one-dimensional Gaussian source assumption, which is commonly used in standard femtoscopic analyses, is not easily applicable. Therefore, a careful assessment of the source geometry and its centrality dependence is required to provide a robust testbed for the interaction models. We have recently joined the STAR collaboration to lead this analysis.

4. Future Prospect: J-PARC-HI

The proposed J-PARC Heavy-Ion (J-PARC-HI) project aims to provide the world's highest intensity heavy-ion beams (beam energy of 1–12 AGeV/c) primarily to explore the QCD phase diagram and search for the critical point at high baryon density. Since the project utilizes a large-acceptance magnetic spectrometer for high-rate hadronic measurements, it offers a versatile platform not only for critical-point searches but also for various studies in hadron and strangeness nuclear physics. As one of such programs, we take an advantage of Λp femtoscopy with spin-polarization observables.

Determining whether the spin-singlet ($^1S_0$) or spin-triplet ($^3S_1$) state exhibits a stronger attraction is a fundamental question in strangeness nuclear physics. While traditional femtoscopy provides pair-spin averaged results, J-PARC-HI will enable a full spin-decomposition. This situation is analogous to scattering experiments where partial-wave

decomposition is achieved through spin observables. In J-PARC-HI femtoscopy, by measuring the proton helicity with a proton polarimeter, we expect to achieve a full spin-decomposition of the Λp correlation function.

By combining these measurements, the spin-singlet and triplet states can be decomposed through the angular correlation between the Λ and proton helicities. In practice, we define a relative azimuthal angle ϕ in the Λp rest frame between the proton emission vector from the $\Lambda \rightarrow p\pi^-$ decay and the vector perpendicular to the $p + C$ scattering plane measured by the proton polarimeter. The $dN/d\phi$ distribution is then expressed as:

$$\frac{dN}{d\phi} \propto 1 - 3\alpha A_C \cos\phi \quad (S_{\Lambda p} = 0, \text{singlet}), \tag{1}$$

$$\frac{dN}{d\phi} \propto 1 + \alpha A_C \cos\phi \quad (S_{\Lambda p} = 1, \text{triplet}), \tag{2}$$

where $\alpha \approx 0.747$ is the Λ decay asymmetry parameter and A_C represents the analyzing power of the $p + C$ reaction in the proton polarimeter, which is approximately 0.3 in the relevant angular range.

The tentative experimental setup for J-PARC-HI femtoscopy is detailed in Ref. [14] (see Fig. 4 in Ref. [14] for the layout). For detailed information on the individual detector components, the reader is referred to Ref. [14]. The setup will incorporate a silicon strip detector (SSD) for vertexing, a large dipole magnet, and a Time Projection Chamber (TPC) for multi-track identification in high-multiplicity events (multiplicity > 100). Particle identification will be further enhanced by a Time-of-Flight (TOF) detector and a forward Ring Imaging Čerenkov (RICH) detector. The proton polarimeter described above will be a key component for these spin-dependent measurements. Fig. 4 shows the expected azimuthal distributions calculated from the updated formulas, where the distinct modulations for singlet and triplet states demonstrate the feasibility of this spin-decomposition technique.

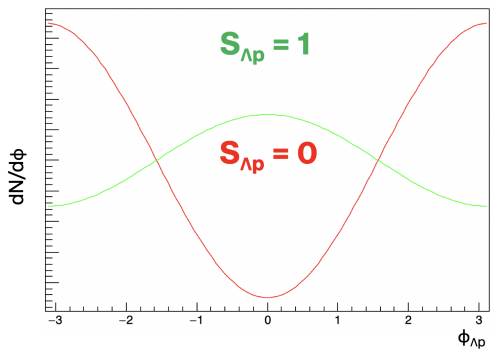

Figure 4: Azimuthal angle distributions for the spin-singlet (red) and spin-triplet (green) states in Λp femtoscopy at J-PARC-HI, calculated according to the updated formulas.

5. Summary

In this paper, we have explored the potential of $\alpha\Lambda$ and Λp femtoscopy at J-PARC and STAR to provide definitive experimental constraints on the short-range part of the ΛN interaction. The $\alpha\Lambda$ system is uniquely suited for this purpose due to its high density and clean spin-isospin state, which avoids the complexities of state decomposition found in other hadronic pairs. The J-PARC E88 experiment, with its primary proton beam, offers an ideal platform to measure these correlations with small source sizes ($\approx$ 1 fm), where the sensitivity to the inner structure of the interaction potential is maximized. Simultaneous analysis of pp, $p\alpha$, and Λp channels will allow for the self-consistent determination of the source sizes, providing a robust experimental foundation. These efforts, complemented by the ongoing analysis of STAR BES data in heavy-ion collisions, are critical for addressing the "hyperon puzzle" and understanding the stiffness of the equation of state in neutron star cores. Looking ahead, the J-PARC-HI project will introduce a revolutionary capability for spin-dependent femtoscopy. By integrating a proton polarimeter and utilizing the weak-decay asymmetry of the Λ, it will be possible to independently determine the scattering parameters for the spin-singlet and triplet states. Such high-precision measurements will be indispensable for discriminating between various theoretical models of baryon-baryon interactions in dense nuclear matter.

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